

Math 150A Section 8 and 16
Spring 2025
Practice Exam II Solutions
March 06, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 10 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	6	
4	10	
5	10	
6	10	
7	7	
Total:	63	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write

(a) if $f''(3) = 0$ then $f(x)$ has an inflection point at 3

(b) the function $f(x) = |x|$ has no critical points

(c) if $f(x)$ is continuous on the interval $[0, 1]$, then it has both absolute maximum and absolute minimum values on $[0, 1]$

(d) if $f'(x) < 0$ for all x , then $f(x)$ can have at most one root

(e) if $f(0) = 0$ and $f(1) = 60$ then $f'(x) = 60$ for some value of x in $(0, 1)$

F,F,T,T,T

2. (10 points) Water is leaking out of an inverted conical tank at a rate of $10,000 \text{ cm}^3/\text{min}$ at the same time water is being pumped into the tank at a constant rate. The tank has a height of 6 m and the diameter at the top is 4 m. If the water level is rising at a rate of $20 \text{ cm}/\text{min}$ when the height of the water is 2 m, what is the rate at which water is being pumped into the tank?

Solution: Always draw a figure first! However, making one on the computer here is inconvenient, so we will skip this step.

Let V be the volume of the tank, h be the height of the water level, and d be the diameter of the water's surface. Then we know $\frac{dh}{dt} = 20 \text{ cm}/\text{min}$ and we want the rate that water goes into the tank. We will get this by calculating the rate of change of the volume in the tank $\frac{dV}{dt}$.

The volume of water in the tank is related to the diameter and the height by

$$V = \frac{1}{12}\pi d^2 h.$$

Also, by similar triangles we know that

$$\frac{d}{h} = \frac{4}{6}.$$

Therefore $d = 2h/3$, and

$$V = \frac{1}{9}\pi h^3.$$

Thus by related rates

$$\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt} = \frac{1}{3}\pi h^2 \cdot 20.$$

At the point when $h = 2$ meters, ie. 200 cm, this says $\frac{dV}{dt} = \frac{800000\pi}{3}$. Now, the change in the volume of the tank is the rate at which water goes in minus the rate at which water goes out, so

$$\frac{800000\pi}{3} = \frac{dV}{dt} = (\text{Rate in}) - (\text{Rate out}) = (\text{Rate in}) - 10000.$$

Therefore

$$(\text{Rate in}) = \frac{800000\pi}{3} + 10000 \text{ cm}^3/\text{min}.$$

3. (6 points) In class, we learned a list of critical properties to consider when sketching a function. Suppose that $f(x)$ is a function which simultaneously satisfies all of the following:
- (A) the domain is $\mathbb{R}$
 - (B) there is a y -intercept at 0; there are x -intercepts at 0 and $\pm 3\sqrt{3}$
 - (C) the function is odd
 - (D) there are no horizontal or vertical asymptotes
 - (E) the function is increasing on $(-\infty, -1)$ and $(1, \infty)$, and is decreasing on $(-1, 1)$
 - (F) it has a local maximum $f(-1) = 2$ and local minimum $f(1) = -2$
 - (G) the function is concave upward on $(0, \infty)$, concave downward on $(-\infty, 0)$ with an inflection point at $x = 0$

Using these properties, we should be able to produce a remarkably good sketch of $f(x)$, even though we don't even know what $f(x)$ is! Create a sketch of $f(x)$ which simultaneously satisfies all the above properties.

Solution: OMITTED

4. (10 points)

(a) State the Mean Value Theorem

(b) State the Extreme Value Theorem

Solution:

(a) MVT says that if $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

(b) The Extreme Value Theorem says that if $f(x)$ is continuous on $[a, b]$ then the absolute maximum and absolute minimum values of $f(x)$ on $[a, b]$ exist.

5. (10 points) For each of the following, provide an example or explain why it does not exist.

(a) a function $f(x)$ with a critical point at 0 but neither a local max nor local min at $x = 0$

(b) a function $f(x)$ which satisfies $f''(1) = 0$ but which does not have an inflection point at $x = 1$

(c) a nonconstant function $f(x)$ whose derivative is always zero

(d) a function with an inflection point at $x = 0$

(e) a differentiable function with three x -intercepts, whose derivative only has one x -intercept

Solution:

(a) $f(x) = x^3$

(b) $f(x) = (x - 1)^4$

(c) can't exist because of MVT

(d) $f(x) = x^3$

(e) can't exist because of MVT

6. (10 points) If $1,200 \text{ cm}^2$ material is available to make a box with a square base and an open top, find the largest possible volume of the box.

Solution: Let x be the width of the box and y be the height. The surface area of the box is $4xy + x^2$, so if we use up all our material we must have

$$4xy + x^2 = 1200.$$

This means

$$y = \frac{300}{x} - \frac{1}{4}x.$$

The volume of the box is

$$V = x^2y,$$

so subbing in the previous equation we get

$$V = 300x - \frac{1}{4}x^3.$$

The critical points of this are $x = \pm 20$, and the absolute max occurs at $x = 20$. At this value, $V = 4000$ so the largest possible volume is 4000 cubic centimeters.

7. (7 points) Consider the function

$$f(x) = x^4 + x^2 - 1$$

- (a) show that $f(x)$ has only one critical point

- (b) explain why the Mean Value Theorem says $f(x)$ cannot have more than two roots

- (c) calculate the value of $f(-1)$, $f(0)$ and $f(1)$

- (d) explain why the Intermediate Value Theorem says $f(x)$ has at least two real roots (and therefore precisely two roots)

Solution:

- (a) $f'(x) = 4x^3 + 2x = 2x(2x^2 + 1)$. The only place where this is zero is $x = 0$, so there is only one critical point.
- (b) If $f(x)$ has three roots at $x = a$, $x = b$, and $x = c$ (in increasing order), then the Mean Value Theorem (or Rolle's Theorem) says that $f'(x) = 0$ at some point between a and b .

Likewise, $f'(x) = 0$ at some point between b and c . However, there's only one point where $f'(x) = 0$, so that's not possible!

(c) $f(-1) = 1$, $f(0) = -1$, and $f(1) = 1$

(d) IVT says that f has a zero between -1 and 0 . Likewise, MVT says f has a zero between 0 and 1 .